capabilities are the key drivers of the consumer electronics market. Here is a sneak-peek of the emerging trends that are likely to find a foothold and secure their role in the consumer electronics sector, now and in the near future.

Connected products through the Internet of Things (IoT). Connectivity and communication are becoming ubiquitous in consumer electronics. This, in turn, has enhanced electronics content as well as the need for technology adoption in this sector in all possible manners of shape, size and application.

For example, flat, rigid sensors and chips may not be a viable option for wearable devices that should comfortably conform to the human body. With the broad and ever-expanding spectrum of smart devices, breakthrough manufacturing technologies must continue to be explored, tested and implemented.

Batteries with enhanced energy storage. Improved battery life is one of the most important deciding factors among consumers. Technological innovations have been continually offering more and more powerful battery technology. Thus, higher energy density batteries will be critical to enabling the kinds of products that deliver the performance consumers want. Although, a significant amount of work has already been done to improve battery technology, many challenges still remain unsolved.

Miniaturisation of devices. With each new generation of devices, consumers expect to receive more and more features in the same size or smaller package along with enhanced quality or safety. This creates a serious need for device and component miniaturisation by pushing the limits of technology—specifically, miniaturisation of next-generation electronic components and systems.

New manufacturing technologies will need to be developed for miniaturised components, including equipment and systems for micro-level automated assembly, micro-sensors and micro-scale electrical connections. Particularly critical will be developing new micro-level assembly techniques and increasing the efficiency of existing techniques.

Flexible and printed electronics. Flexible and printed electronics offer flexible circuitry that enables manufacturers to create consumer electronics products with unconventional functionality and design. Displays, sensors and printed memory can be bent, wrapped, rolled and stretched out to a great extent, resulting in reduction of cost, weight and power consumption of electronic devices.

Further applications are also possible for product development cycle optimisation with the advent of technology.

3D and 4D printing. 3D printing or additive manufacturing has widespread useful applications in the consumer electronics industry, especially in the printing of electronic circuits. The ability to print circuitry offers immense flexibility in design-